A study of comparison of various parametric statistical methods used for analysis of microarray data

Journal:	*Statistical Methods in Medical Research*
Manuscript ID	SMM-18-0411
Manuscript Type:	Review Article
Keywords:	liver, liver cancer, differentially expressed genes, parametric statistical methods, microarray
Abstract:	Microarray experiments produces huge amounts of expression measurements of large number of genes across different conditions. Microarray data analysis evolved as an advanced research topic during recent decades. Microarray expression data is analysed to predict differentially expressed gene, many types of statistical methods have been developed and modified accordingly. The main reason for doing analysis on microarray data is to select and predict the genes which shows different expression values under different experimental conditions. Aim of this research paper is to explore various types of parametric statistical methods proposed to analyse microarray expression data for identifying the differentially expressed genes, and to investigate the mechanism of disease development. Besides, we also predicted the best condition for each method where they perform better and a comparative analysis of various methods has been done. Many kinds of parametric statistical methods have been studied to identify and select differentially expressed genes, only very few studies have compare the performance of these methods. In our study, we extensively study and compare the different types of parametric methods by using the simulated datasets.

A study of comparison of various parametric statistical methods used for analysis of microarray data

Meenu Sharma Department of Computer Science, Jamia Millia Islamia, New Delhi, India and
Rafat Parveen Department of Computer Science, Jamia Millia Islamia, New Delhi, India

Microarray experiments produces huge amounts of expression measurements of large number of genes across different conditions. Microarray data analysis evolved as an advanced research topic during recent decades. Microarray expression data is analysed to predict differentially expressed gene, many types of statistical methods have been developed and modified accordingly. The main reason for doing analysis on microarray data is to select and predict the genes which shows different expression values under different experimental conditions. Aim of this research paper is to explore various types of parametric statistical methods proposed to analyse microarray expression data for identifying the differentially expressed genes, and to investigate the mechanism of disease development. Besides, we also predicted the best condition for each method where they perform better and a comparative analysis of various methods has been done. Many kinds of parametric statistical methods have been studied to identify and select differentially expressed genes, only very few studies have compare the performance of these methods. In our study, we extensively study and compare the different types of parametric methods by using the simulated datasets.

1 Introduction

It provides a way in biotechnology, to analyse the behaviour of thousands of gene on the basis of their expression profiles simultaneously by the use of hybridization technique.[1]On the basis of amount of hybridization, each microarray spot has its unique fluorescence intensity. By using this fluorescence intensity, measurement of the level of gene expression of a sample is done. This type of measuring gene expression level by using fluorescence intensity has limitations with respect to reproducibility and resolution.[2] Expression profiles of genes study is of relevance because it provides a way to identify or to predict disease markers that are having relevance in medical treatments. Many researchers want to identify genes which are differentially expressed in diseased condition .[3]

Address for correspondence: Meenu Sharma, Department of Computer Science, Jamia Millia Islamia, New Delhi 110025, India.
E-mail: mshwetab@gmail.com

Past many years, various statistical tests have been developed to do microarray data analysis. Numerous statistical tests with their used approach is shown in Table 1.These all methods classified into two categories such as parametric methods and non-parametric methods, and their comparison is presented in Table 2. The classification of all the methods is shown in Fig. 2.Main aim of this comparison is to screen out the best methods that exactly identify the highest proportion of differential expressed genes with having fewer fractions of equivalently expressed (EE) genes which are mistakenlycountedin differential expressed genes.[3]Researchers are able to identify only a small fraction of differentially expressed genes, by using statistical methods and it is the major problem during the selection of genes. Differentially expressed genes are called significant or discriminative genes.[6]

Numerous latest articles [1,6–10] are present which has provided the comparative study of different variety of statistical methods which works on microarray data. Moreover, in the above-mentioned papers, either they have been reviewed a large number of statistical methods neglecting its individual special phenomenon or they covered few tests with focus on the specific phenomena for every statistical test. But these both aspects are not covered under a single study. Due to this reason, we tried to include both these aspects, in this paper. Here, we have shown anextensive study of numerous parametric and non-parametric statistical testing [1, 3,5–15]methodologies used for predicting genes which shows differentially expressed pattern from microarray data sets. At last, a list of merits and demerits of each method are provided, with their directions under which further investigation needed. Similar conclusion is drawn by Kim et al.[1]as Jeffery et al.[16]like the sample size, data distribution, data variance, genes number, number of samples etc. for each test has impact on which test performs better.

In this paper, we compare various parametric methods,like T-statistics, B-statistics and Bayes T-test or Bayesian methods, ANNOVA-1 AND Pearson Correlation Test, three different methods come under categories other than parametric and non-parametric methods such as Fold change, RAM and Rank product. In section 2, a review details are presented on each of these various statistical tests used for identifying genes which shows differential expressedpatterns under different experimental conditions. A comparative study with analysis of various statistical methods shown in Section 3. Statistical methods are employed on data sets which are explained in Section 4, and their results with discussion on different statistical methods are explained in Section 5. Finally, conclusion is presented.

2 Various statistical tests for predicting differentially expressed genes

2.1 Fold Change

It is the most popular method because of its simplicity. In literature, two definitions for fold change are found. Standard definition [17] is the ratio of the means under two different experimental conditions.

$$FC_i = \frac{\overline{x}_{i1}}{\overline{y}_{i2}}$$

Where, $\overline{x}_{i1}$represents the mean expression of gene i in j number of replicates under experimental condition 1 (i.e. diseased or treated), $\overline{y}_{i2}$represents the mean expression of gene i in k number of replicates under experimental condition 2 (i.e. control). As stated in [18] and [19], the fold change for gene i can be defined as the difference between the means specified to two-different experimental condition or samples such as diseased/treated group and the other is for the control group. This is work especially

Table 1 Various methods developed to analyse microarray gene expression profile with their approach

Sr No	Statistical tests	Used Approach
1	Fold change	Rules on the fold increase/decrease cut-off
2	T-statistics	Permutation Approach
3	Bayes t-test or Bayesian Methods	Parametric Bayesian method
4	B-statistics	Empirical Bayes Approach
5	ANOVA 1 test	Permutation Approach
6	Pearson's Correlation Test (corr)	Pearson's Correlation Coefficient
7	SAM	Statistic similar to t-statistics
8	SAMROC	Permutation Approach
9	Zhao-Pan method	Null distribution of test statistic
10	SDEGRE	Permutation approach based on relative entropy and kernel density estimation
11	Permutated T-test (perm)	A kind of t-test uses permutation or a rearrangement
12	Wilcoxon Ranksum Test (RST)	Based on the difference of mean Ranks, rank based
13	Modified Wilcoxon Ranksum Test (MRST or ModRST)	Based on the difference of mean Ranks, rank based
14	LIMMA	Uses matrix expression data, then fits a linear model to each row of data
15	Shrink-t	James-Stein ensemble shrinkage estimation rule
16	Soft-Threshold aka L1 penalized t-statistic	Uses t statistics
17	Rank product	Fold change
18	Ranking analysis of Microarray	Standard permutation procedure for each gene

Table 2 Brief comparison of parametric and non-parametric statistical test

Sr No	Parametric methods	Non-parametric methods
1	Assumes that data follows normal distribution	Do not assume any particular distribution, mainly assumes data follows a non-normal distributed data
2	Same variance property of the group	NA
3	Relation to data is expressed in absolute numbers instead of ranks	It has the ability to relate data expressed in ranks rather than the absolute numbers of values
4	Normal distributed data	Non-normal distributed data
5	Strong normality assumption	Not having any strong normality assumption
6	Assuming null hypothesis of showing no genes, which are differentially expressed	Computation of null distribution of the test statistic instead of considering null hypothesis of no differential gene expression
7	Used sample data of population which should follow a probability distribution, based on prefixed parameters set	Sample data comes from a given probability distribution, it is the opposite to the parametric statistics
8	More powerful than non-parametric tests	Less powerful as compared to parametric tests
9	Not perform best with unexpected outlying observations as compared to non-parametric approach	Useful for dealing with unexpected outlying observations, that might be problematic with a parametric approach

when all values are transformed in $log2$scale.

$$FC_i = \overline{x}_{i1} - \overline{y}_{i2}$$

We call these variety of fold change as FC_{ratio} and $FC_{difference}$, respectively. Fold change method is not performed well in selectinggenes which shows different expression in microarray dataset, mainly under less abundant transcripts.

2.2 Parametric Tests

Classical test is the other popular name used for parametric tests.These statistical tests belong to a branch of statistics which consider that sample data follows a normal distribution and same variance within the groups. If these expectations are not authentic, then non-parametric statistical tests must be used[20]. Various types of parametric statistical tests are developed,which are utilized for identifying the differentially expressed genes, are briefly explained below.

2.2.1 T-Statistics

The most basic and fundamental statistical tests frequently used for identifying differentially expressed genes is two sample t-statistic[9].Let us consider, $\overline{x}_{i1}$ and $\overline{y}_{i2}$ are two independent data of gene expression under two different experimental conditions 1 and 2, such as normal or diseased respectively for a specific gene i, so the two-sample t-test is computed as below:

$$t = \frac{(\overline{x}_{i1} - \overline{y}_{i2})}{s_i}$$

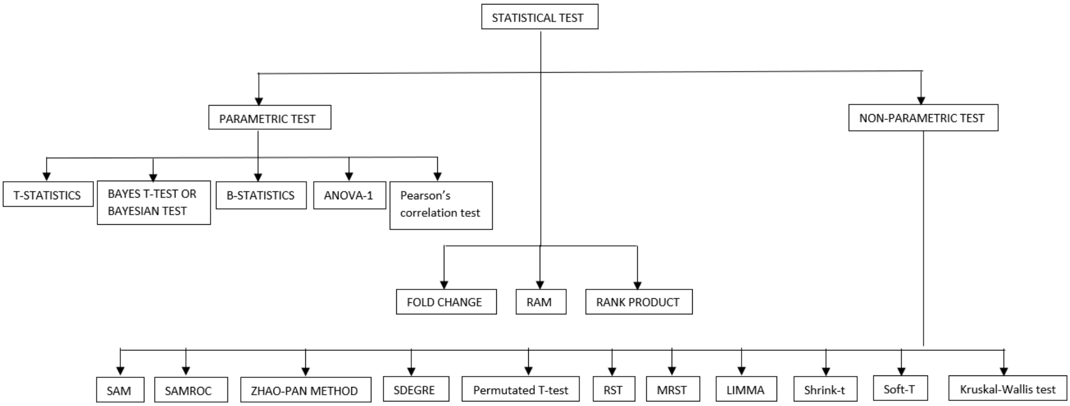

Figure 1 Classification of different types of statistical test

$$s_i = s * \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

Here $$s = \sqrt{\frac{(n_1-1)*s_{i1}^2+(n_2-1)*s_{i2}^2}{(n_1+n_2-2)}}$$

Where, n_1 and n_2 represents the samples size in experimental condition 1 and 2 respectively, and s represents the pooled sample variance. Degree of freedom (df) is used in pooled sample variance formula in the form of $(n_1 + n_2 - 2)$, s_{i1} and s_{i2} used to represents the standard deviation of replicated data in experimental conditions 1 and 2 respectively while s_{i1}^2 and s_{i2}^2shows their variance, and their variance is equal. This assumption of equal variability is known as "Behrens Fisher" problem[21]. It is required to know whether the variance under two experimental conditions is equal or not. Under the equal variance situation, used the earlier t-statistics mentioned above otherwise use the t-statistics given below.

$$t = \frac{(\overline{x}_{i1} - \overline{y}_{i2})}{\sqrt{\frac{s_{i1}^2}{n_1} - \frac{s_{i2}^2}{n_2}}}$$

Here, un-pooled standard deviation is used. It is known as Welch's test[21].It is somehow similar to the t-statistic, but the only difference is that a known variance is used in denominator.

2.2.2 *Bayes t-test or Bayesian method*

For analyzing microarray data, a Bayesian probabilistic framework is developed by Baldi and Long[27]. This statistic is utilized to resolve the problem arises due to small variance under the less expressed values and utilizes the Bayesian statistical test in parametric form to estimate the parameters value such as standard deviation, mean etc. for T statistical test. It is more preferable and popular in analyzing the small sample size, because of its effectiveness under small sample size data. Bayes T-test utilizes the characteristics like mean (μ) and variance (σ^2) estimated by Bayesian method rather by T-statistic. Posterior estimate mean for each group is as follows

$$\mu_k = \mu_{nk}, \sigma_k^2 = \frac{v_k \sigma_{nk}^2}{v_k - 2}$$

Here, μ_{nk} is the posterior estimate mean and a convex weighted average of the prior mean μ_{0k} and $\bar{p}_k$ is the sample mean for group $k, k = 1,2$, such as

$$\mu_{nk} = \frac{\lambda_{0k}}{\lambda_{0k} + m_k} \mu_{0k} + \frac{m_k}{\lambda_{0k} + m_k} \bar{p}_k$$

Here μ_{0k} and $\frac{\sigma_{0k}^2}{\lambda_{0k}}$ are the hyperparameters, can be described as the position and the scale of μ_j, respectively, sample size of each group is represented by m_k. Posterior variance component is shown byσ_{nk}^2.

2.2.3 *B-statistics*

A statistic is proposed by Lonnstedt and Speed[14] made on the basis of empirical Bayes approach by utilizing a Bayes log posterior odds. This statistic overcome T-statistic problem arises by log odds posterior statistic undernormality assumption. Lonnstedt and Speed hierarchical model is implemented into a practical approach by Smyth[12] and named it B-Statistic. On the basis of general linear model, model is reset with random coordinates and contrast of interest. B-statistic made by Smyth's is shown as follows:

$$B = \frac{p(\beta_i \neq 0|\tilde{t}_i, s_i^2)}{p(\beta_i = 0|\tilde{t}_i, s_i^2)} = \frac{p}{1-p}\left\{\frac{v_i}{v_i+v_0}\right\}^{1/2} \cdot \left\{\frac{\tilde{t}_i^2 + d_0 + d_i}{\tilde{t}_i^2\left(\frac{v_i}{v_i+v_0}\right) + d_0 + d_i}\right\}^{(1+d_0+d_i)/2}$$

Here, β_i is used to represents the regression coefficient of contrast of interest, and $\hat{\beta}_i$ is used as contrast estimator which is expected as roughly around normal with mean β_i and matrix of covariance $C^T V_i C \sigma_i^2$. Where, V_i is the unscaled covariance matrix, s_i^2 represents the residual sample standard deviation and gene i has linear model and whose residual degree of freedom has been represented by d_i. Prime estimators of v_i and d_i are represented by v_0 and d_0 respectively. p is the differentially expressed genes expected proportion.

Smyth's developed a 'moderated' T-statistic as follows

$$\tilde{t}_i = \frac{\hat{\beta}_i}{\tilde{s}_i\sqrt{v_i}}, \tilde{s}_i^2 = \frac{d_0 s_0^2 + d_i s_i^2}{d_0 + d_i}$$

Here, $\hat{\beta}_i$ represented the regression coefficient estimate for the contrast of interest. B-statistics will depend on hyperparameters v_0, d_0, s_0^2 and p, which can be calculated from the data.

2.2.4 ANOVA-1 test

ANOVA-1(Analysis Of Variance-1)[9] is a statistical test in which means of various groups or samples at different experimental conditions may or may not be equal for all genes due to this reason it is concluded that t-test work on greater than two groups. Performing this multiple two sample t-test result in a more probability of making type-1 error[22, 23]. So due to this, ANOVA-1 test is more effective when we want to make comparison between means of two or more than two populations/groups. This method compares the means of two or more groups/population by comparing the estimated variance. The aim of doing this variance comparison is to know the mean difference between groups for creating statistical significance by analyzing variance. The basic of ANOVA-1 is that the two-independent estimation of population variance which computed from the sample dataset. Then, proportion is calculated for those two evaluates, represented by F. Let us consider, for experimental condition 1: number of treated samples aret_1, sum of all the values of group 1= V_1, mean is $\bar{X}_1$ and for experimental condition 2: number of controlled samples are t_2, sum of all the values of group 2= V_2, mean is $\bar{X}_2$. Total number of groups = G (in our case, here G=2), total number of values in these two-experimental condition $T = \sum_{j=1}^{G} t_j$, and sum of all the values under all experimental conditions $V = \sum_{j=1}^{G} V_j$. Suppose$C_1 = \frac{V^2}{T}$, $C_2 = \sum_{j=1}^{G}(\sum_{i=1}^{t_j} x_{ij}^2)$ here, x_{ij}^2 is the value of i gene in j number of replicates, and$C_3 = \sum_{j=1}^{G} \frac{V_j^2}{t_j}$. Thus, sum of squares between groups$SS_{between} = C_3 - C_2$, sum of squares within groups$SS_{within} = C_2 - C_3$, and total sum of squares$SS_{total} = C_2 - C_1$. Now dof between groups $dof_{between} = (G-1)$, and dof within groups $dof_{within} = \sum_{j=1}^{G}(t_j - 1)$. Here, $dof_{total} = \sum_{j=1}^{G} t_j - 1$,

$$var_{between} = MeanSquare_{between}(MS_{between}) = \frac{SS_{between}}{dof_{between}}$$

$$var_{within} = MeanSquare_{within}(MS_{within}) = \frac{SS_{within}}{dof_{within}}$$

$$F = \frac{MS_{between}}{MS_{within}}$$

2.2.5 Pearsons's Correlation Test (corr)

Correlation is a statistical measure of extent to two variables varied with respect to each other; it can be positively or negatively correlated. In correlation, assumption is made that data is from the bivariate population (involves two variables). Pearson correlation coefficient[24] is a way to represent the strength of linear association found between two variables. Let us consider two groups, group 1 has P number of samples, with expression values as t, with mean $\bar{t}$and standard deviations_t, and other group 2, with expression values v, having mean $\bar{v}$ and standard deviation s_v. Pearson's correlation coefficient (usually represented by ρ) of two variables can be represented as the fraction of the covariance of both the variables to the product of their standard deviations. Such as

$$\rho = \frac{cov\,(t, v)}{s_t s_v}$$

where $$cov(t, v) = \sum_{i=1}^{P}(t_i - \bar{t})(v_i - \bar{v})$$

This test is used to identify whether two variables are correlated. This test is not able to predict that two variables are not related to each other. T-statistics can be calculated for correlation coefficient (ρ) such as

$$t = \frac{\rho}{s_\rho}$$

Where $$s_\rho = \sqrt{\frac{1-(\rho)^2}{P-2}} \text{ and}$$

$$dof = P - 2$$

There is only one dof for each group.

2.3 Rank Product

The rank product[3] method is made to overcome the problem arises in fold change method, it is statistically both at the same time hard as well as simple[25]. After gaining the recognition, as a statistical test for identify genes which are differentially expressed from microarray data, Koziol[26] defines it in an extended form onto two sample data settings. Koziol defines this statistic method as follows:

$$\mathrm{RP_i} = \prod_{m=1}^{M} \mathrm{R}_{\mathrm{im}}^{\frac{1}{M}} \div \prod_{n=1}^{N} \mathrm{R}_{\mathrm{in}}^{\frac{1}{N}}$$

Here, M and N are used to represents the number of replicates under the experimental condition 1 and 2 respectively, and rank is assigned among the expression of n genes inside a single sample, and it is performed for each sample. $\mathrm{R_{im}}$ and $\mathrm{R_{in}}$is for showing the ith gene ranks under experimental condition 1 and 2 respectively. Further, on this test statistic the montone log transformation is appliedto get the better approximation of the test statistic and null distribution, which is shown below:

$$\log \mathrm{RP_i} = \frac{1}{M}\sum_{m=1}^{M} \log \mathrm{R_{im}} - \frac{1}{N}\sum_{n=1}^{N} \log \mathrm{R_{in}}$$

As stated by Koziol[26], "the exact distribution of log(RPi) can be tedious" so for this distribution a normal approximation is sufficient, mainly for large samples size. This approximation is not performed well, if data has skewed.

2.4 Ranking analysis of Microarray (RAM)

RAMis a large scale two sample t-statistic[15], works on the basis of comparison made between ranked t-statistics set and ranked Z-values (one set is produced from ranked estimated null scores). These ranked Z-values are estimated using "random splitting" approach instead of permutation approach. RAM performs well under undesirable conditions like small sample size, large fudge factor and data containing numerous distribution of noises than SAM[15].

This methodology involves the ranking of genes, based on their value calculated by some selected statisticalmethods, in our case is T. Let consider, $T_{i^*}$is the largest value of T at i^{th} position, it means that this i genes is said to have significant different expression levels under two experimental conditions for a threshold value Δ which is given, if

$$|T_{i^*} - Z_{i^*}| > \Delta,$$

Here, $Z_{i^*} = E(T_{i^*})$ stands for the expectation of Z_{i^*} under the condition for null hypothesis such that there is no gene who has significant difference between two experimental conditions.To estimate Z value, first randomly split the sample into two subpartsnot having size difference more than a predefined C value. They found that it is acceptable to utilize four value for C. For K^{th} split, suppose $\bar{s}^K_{spi}$ is the mean of subsample s, sample p for gene i. Define $\bar{e}^K_{1i} = \bar{x}^K_{11i} - \bar{x}^K_{12i}$ and $\bar{e}^K_{2i} = \bar{x}^K_{21i} - \bar{x}^K_{22i}$, and hence,

$$\bar{e}^K_i = \frac{1}{2}(e^K_{1i} - \bar{e}^K_{2i})$$

This splitting process is done for each and every gene, then define $Z^K_i = \frac{e^K_i}{\delta_i}$. The estimated set of values of Z^K_i are ranked. Consider, $Z^K_{i^*}$ is the largest value in the set of K-th split. Further, estimate the $Z^K_{i^*}$ by using the mean of $Z^K_{i^*}$ across all the splits, shows below

$$\bar{Z}^K_{i^*} = \frac{1}{N}\sum_{K=1}^{N} Z^K_{i^*}$$

3 Theoretical comparison

Remarkable improvement has taken place in the statistical tests used to study the microarray expression data with respect to detection of differentially expressed genes. After analysing all the statistical methods which are used to analyse microarray data, except non-parametric methods, a comparative Table 3 is prepared, which gives the summary of applied statistical methods, with their best performance condition, advantages and limitations.

Table 3 Comparison between different parametric statistical tests used to analyse microarray data.

Test/parameter	Sample size	Data distribution	Variance	Advantages	Limitations
Fold change	Small	Both (normally and non-normally distributed)	Do not control variance	Popular because of its simplicity and easy to understand	This method not take variability into consideration
T-statistics	Large	Normally	Same or	Popular because of its	More error prone (high

			equal variance	simplicity and easy to understand	FDR) because of variance problem arises due to low level of genes expression value.
Welch's t	Large	Normally	Unequal variances of two groups	More robust than t-test; This test is simpler to use than other two-sample tests with comparable statistical power.	This test should not be used on lognormal data. Under skewed data distribution and small sample size, authenticity decreases.
Bayes t-test or Bayesian methods	Small	Both	Assume unequal variance	Popular because of its effectiveness in evaluating samples size dataset, it is depending upon the parametric assumptions	We need a value for prior proportion of differentially expressed genes, the ranking of genes by their B value change only marginally for genes with large B when the parameter estimates are altered, the actual scale changes. Thus, we cannot rely on any standard cutoff value, such as B=0, for the selection of differentially expressed genes
B-statistics	Large	Non-normal	Require equal variance	Popular because it analyze small sample size effectively, B-statistics is depending upon hyperparameter and these parameters are depending on the data.	Difficult to estimate hyperparameters, the main problem in computing c, involves it considers only the distribution of genes which are differentially expressed.
ANOVA1 test	Large	Normal	Same variance	Useful under the situation where comparison is made in mean between two or more groups.	Similar performance with t-test, compare means of 3 or more groups, Inappropriate if the different columns represent different variables instead of different groups
Pearson's Correlation Test (corr)	Poor in both sample size	Poorer than others for both types of distribution	Do not control variance	Commonly used in linear regression	Not applicable for predicting differentially expressed gene/miRNA except certain situations, valid for linear relationship between the variables
Rank product	Small	Poor performance in skewed data distribution	Constant variance	Statistically simple and rigorous at the same time	Does not perform will if data is skewed
RAM	All sample size	Gamma and normally distribution	NA	Insensitive to the treatment effect that often presents in real data and having a better estimate of FDR It works well for small sample sizes which is particularly useful for analyzing microarray data that often have small sample sizes.	Less number of permutated samples are needed for small sample size, which result to a biased ranking test and even gives the test not applicable.

4 Experimental datasets

For performing experiments, we have generated simulated datasets are of 10 sample sizes having 1,000 genes under both the experimental conditions such as diseased or normal, assumes the same variance (such as std1=std2=2) but with different means (mean1 = mean2 + 2). Datasets which follow Normal distribution utilized for both the conditions. We selected normal distribution patterns not only because they will generate the homogeneity/symmetry in data distribution for each condition, but also because in this research paper comparison includes the parametric statistical tests only.

5 Simulation results and discussion

Our aim is to find differentially expressed genes accurately[28,29], for this purpose we used Matlab and R software's. Simulated datasets have been prepared for performing simulation. Datasets comprises of rows which represents genes and columns representing the samples. Then used filtering procedure to remove those genes who have low variance, for this purpose we used 'genevarfilter' Matlab function. Then we have utilized the parametric statistical tests, because these tests are well suited for normally distributed datasets. For performing one way ANOVA and pearson correlation coefficient statistical tests, Matlab software package is employed for performing t-test while R Bioconductor package is employed for performing Welch's t-test. Corresponding p-value has been calculated for each and every gene by applying each one of statistical test. After that, Volcanoplot are utilized for predicting differentially expressed genes according to their ranks.

An extensive simulation study has been done for evaluating the performance of each and every earlier method.We generated 1000 genes of different sample sizes (10, 20 and 40 samples) data. Fig.3 shows the simulation results and Fig. 4, Fig. 5 and Fig. 6 shows the Volcanoplots having equal variance and different means. In normally distributed data T-statistics, one way ANOVA and Welch's t-test together perform well, rather Pearson's correlation test not perform well in all the sample sizes but it perform best with small sample size as compared to large sample size.

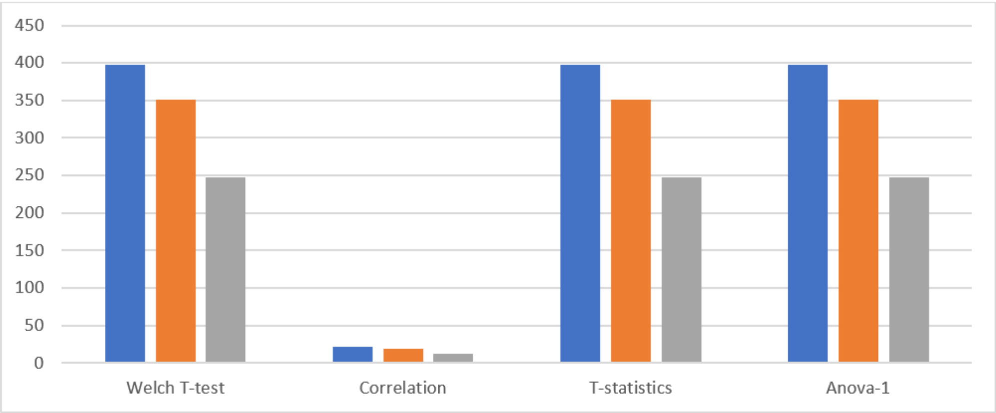

Figure 2 Differentially expressed genes under different statistical test: In this Figure, Blue color represent the results comes in sample size 10, Orange color represent the results comes in sample size 20 and Grey color represent the results comes in sample size 40.

After performing experiments, some useful information is extracted which are explained as Normality test should be applied before employing any tests for selecting the genes having different patterns of expression under two experimental conditions, mainly for small sample size. Welch T-test, T-test and Annova1 are all performing good under all conditions of small and large sample sizes, but Correlation is performing the worst under all cases (sample sizes) for simulated dataset. The performances of Welch T-test, t-test and ANOVA 1 are quite similar as can be seen from Fig. 4, Fig. 5, and Fig. 6.Figs. 4b, 5b and 6b signifies the faulty performance of Correlation test for all sample sizes for the simulated data sets. To summarize, Pearson’s correlation tests are fundamentally not the correct statistical tests for predicting differentially expressed genes.

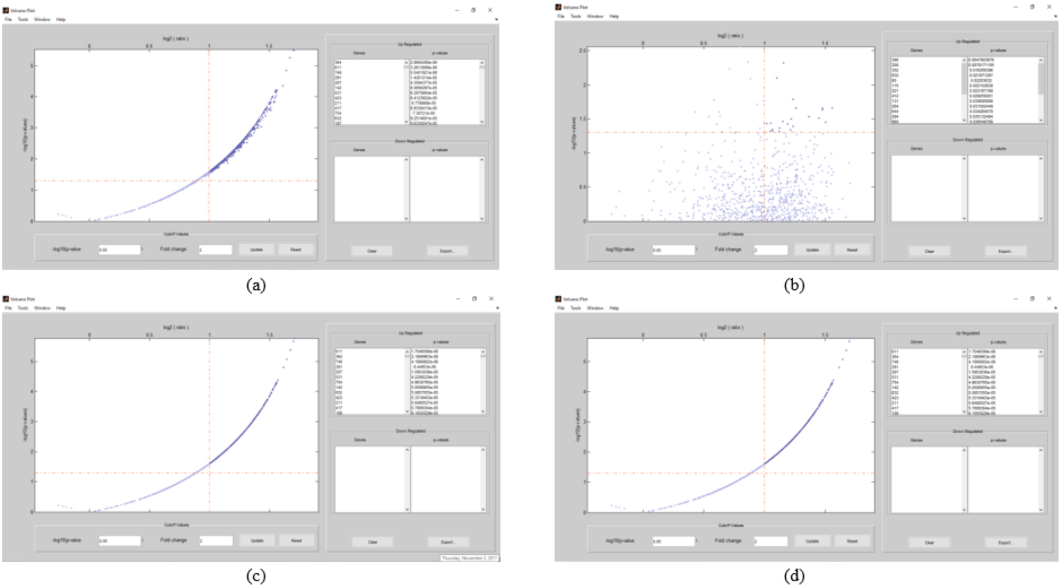

Figure3 Volcanoplot of predicted differential up and down regulated genes from the simulated datasets of sample size 10 by different types of statistical tests -(a) Welch T-test, (b) Correlation, (c) T-test, and (d) Annova1.

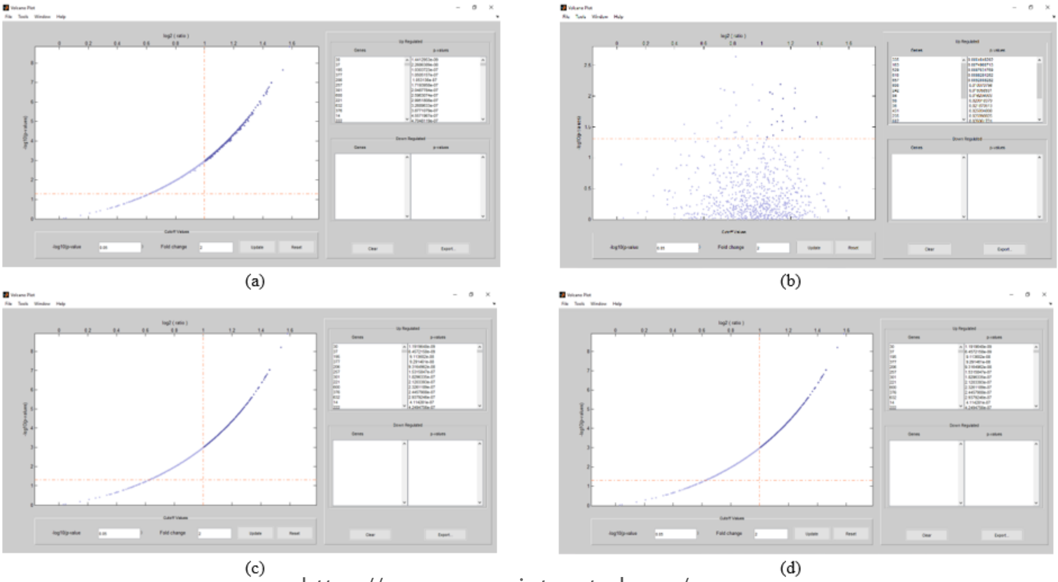

Figure 4 Volcanoplot of predicted differential up and down regulated genes from the simulated datasets of sample size 20 by different types of statistical tests -(a) Welch T-test, (b) Correlation, (c) T-test, and (d) Annova1.

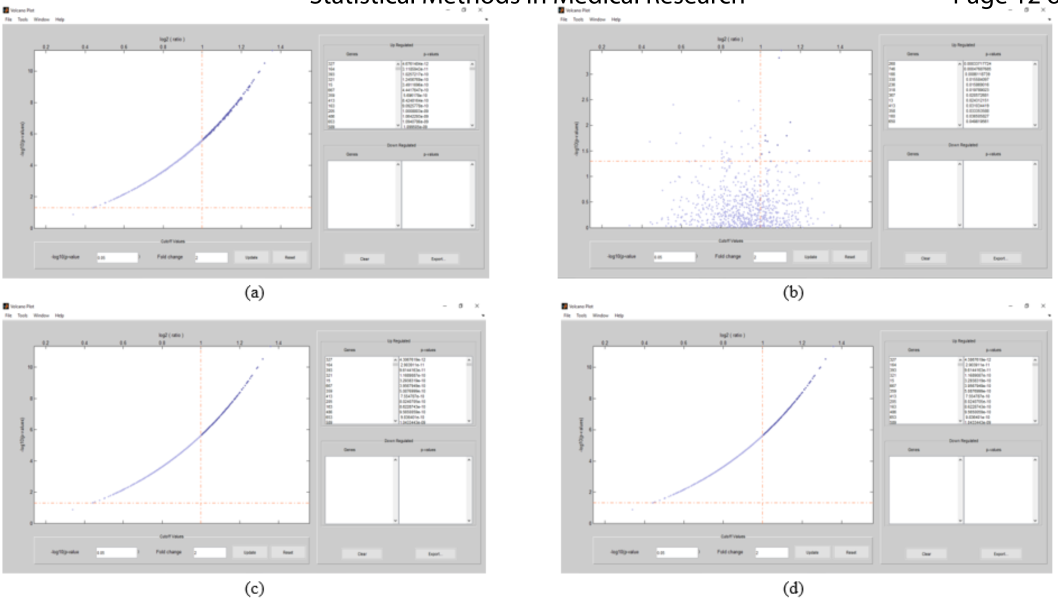

Figure 5 Volcanoplot of predicted differential up and down regulated genes from the simulated datasets of sample size 40 by different types of statistical tests -(a) Welch T-test, (b) Correlation, (c) T-test, and (d) Annova1.

If data comprised of two normally distributed populations having same variances, the two-sample t-test, welch t-test and Anova1are more powerful than Pearson correlation test.

5 Conclusion and Discussion

An extensive review of numerous parametric statistical tests has been provided in this paper, which are used to predict genes having different pattern of expression under two experimental conditions. These parametric methods are evaluated by performing experiments by assuming p-value cutoff of 0.05 for the three simulated datasets. Statistical tests performance has been affected by few conditions like size of sample, data distribution, variance of data, genes number, etc. So, for getting the reliable and significant results, we first require to get information about the real characteristics of the given dataset and then employ the correct appropriate statistical method on that particular dataset. The traditional statistical tests applied on microarray data to identify differentially expressed genes, or a clustering algorithm to identify the groups of genes that shows similar behaviour. This methodology is not successful for predicting the genes co-expressed differentially[30] which have significant biological roles. Statistical test which have better performance to identify genes which shows co-expressed differential patterns are not many, for this reason further work is needed in this field.

Future work

This work was supported by Korea Research Foundation Grant (KRF-2002-075-C00005). JW Lee was also supported by Korea Science and Engineering Foundation Grant (R14-2003-002-01002-0).

References

1 Kim SY, Lee JW, Sohn IS. Comparison of various statistical methods for identifying differential gene expression in replicated microarray data. Stat Methods Med Res. 2006;15(1):3-20. doi:10.1191/0962280206sm423oa.

2 An J, Kim K, Kim S. An algorithm for identifying differentially expressed genes in multiclass RNA-seq samples. 2014 Int Conf Big Data Smart Comput BIGCOMP 2014. 2014:40-44. doi:10.1109/BIGCOMP.2014.6741402.

3 Andrew H, Florence G, Bm GK. Methods for Identifying Differentially Expressed Genes: An Empirical Comparison. 2015;6(5). doi:10.4172/2155-6180.10002.

4 Dudoit S, Yang YH, Callow MJ, Speed TP. Statistical methods for identifying differentially expressed genes in replicated cDNA microarray experiments. Stat Sin. 2002;12(August):111-139. doi:10.1146/annurev.psych.53.100901.135153.

5 Yan X, Deng M, Fung WK, Qian M. Detecting differentially expressed genes by relative entropy. J Theor Biol. 2005;234(3):395-402. doi:10.1016/j.jtbi.2004.11.039.

6 Bandyopadhyay S, Mallik S, Mukhopadhyay A. A Survey and Comparative Study of Statistical Tests for Identifying Differential Expression from Microarray Data. IEEE/ACM Trans ComputBiolBioinform. 2013;11(1):95-115. doi:F45658F6-CA2A-4A82-A790-B4B467E35BE2.

7 Pan W. A comparative review of statistical methods for discovering differentially expressed genes in replicated microarray experiments. Bioinformatics. 2002;18(4):546-554. doi:10.1093/bioinformatics/18.4.546.

8 Vickers AJ. Parametric versus non-parametric statistics in the analysis of randomized trials with non-normally distributed data. BMC Med Res Methodol. 2005;5(1):35. doi:10.1186/1471-2288-5-35.

9 Sreekumar J, Jose K. Statistical tests for identification of differentially expressed genes in cDNA microarray experiments. Indian J Biotechnol. 2008;7(October):423-436. http://nopr.niscair.res.in/handle/123456789/2356.

10 Murie C, Woody O, Lee AY, Nadon R. Comparison of small n statistical tests of differential expression applied to microarrays. BMC Bioinformatics. 2009;10(1):45. doi:10.1186/1471-2105-10-45.

11 Cui X, Hwang JTG, Qiu J, Blades NJ, Churchill GA. Improved statistical tests for differential gene expression by shrinking variance component estimates. Biostatistics. 2005;6:59-75.

12 Smyth GK. Statistical Applications in Genetics and Molecular Biology Linear Models and Empirical Bayes Methods for Assessing Differential Expression in Microarray Experiments. 2011;3:2-3. doi:10.2202/1544-6115.1027.

13 ElBakry O, Ahmad MO, Swamy MNS. Identification of differentially expressed genes for time-course microarray data based on modified RM ANOVA. IEEE/ACM Trans ComputBiolBioinform. 2012;9(2):451-466. doi:10.1109/TCBB.2011.65.

14 Lonnstedt I, Speed T. Replicated microarray data. Stat Sin. 2002;12(12):31-46.

15 Tan Y-D, Fornage M, Fu Y-X. Ranking analysis of microarray data: a powerful method for identifying differentially expressed genes. Genomics. 2006;88(6):846-854. doi:10.1016/j.ygeno.2006.08.003.

16 Jeffery IB, Higgins DG, Culhane AC. Comparison and evaluation of methods for generating differentially expressed gene lists from microarray data. BMC Bioinformatics. 2006;7:359. doi:10.1186/1471-2105-7-359.

17 Tusher VG, Tibshirani R, Chu G. Significance analysis of microarrays applied to the ionizing radiation response. Proc Natl Acad Sci. 2001;98(9):5116-5121. doi:10.1073/pnas.091062498.

18 Guo L, Lobenhofer EK, Wang C, et al. Rat toxicogenomic study reveals analytical consistency across microarray platforms. Nat

Biotechnol. 2006;24(9):1162-1169. doi:10.1038/nbt1238.

19 Choe SE, Boutros M, Michelson AM, Church GM, Halfon MS. Preferred analysis methods for Affymetrix GeneChips revealed by a wholly defined control dataset. Genome Biol. 2005;6(2):R16. doi:10.1186/gb-2005-6-2-r16.

20 Chen YL, Hu HL. An overlapping cluster algorithm to provide non-exhaustive clustering. Eur J Oper Res. 2006;173(3):762-780. doi:10.1016/j.ejor.2005.06.056.

21 Weir JBD V. Significance of the difference between two means when the population variances may be unequal. Nature.1960;187(4735):438. doi:10.1038/187438a0.

22 Mccluskey A, Lalkhen AG. Statistics IV: Interpreting the results of statistical tests. Contin Educ Anaesthesia, Crit Care Pain. 2007;7(6):208-212. doi:10.1093/bjaceaccp/mkm042.

23 Rothman KJ. Curbing type i and type II errors. Eur J Epidemiol. 2010;25(4):223-224. doi:10.1007/s10654-010-9437-5.

24 Aldrich J. Correlations Genuine and Spurious in Pearson and Yule. Stat Sci. 1995;10(4):364-376. doi:10.2307/2246135.

25 Breitling R, Armengaud P, Amtmann A, Herzyk P. Rank products: A simple, yet powerful, new method to detect differentially regulated genes in replicated microarray experiments. FEBS Lett. 2004;573(1-3):83-92. doi:10.1016/j.febslet.2004.07.055.

26 Bee MA. NIH Public Access. 2009;75(5):1781-1791. doi:10.1007/s11103-011-9767-z.Plastid.

27 Baldi P, Long AD. A Bayesian framework for the analysis of microarray expression data: regularized t -test and statistical inferences of gene changes. Bioinformatics. 2001;17(6):509-519. doi:10.1093/bioinformatics/17.6.509.

28 Leek JT, Scharpf RB, Bravo HC, et al. Tackling the widespread and critical impact of batch effects in high-throughput data. Nat Rev Genet. 2010;11(10):733-739. doi:10.1038/nrg2825.

29 Scherer A. Batch Effects and Noise in Microarray Experiments: Sources and Solutions.; 2009. doi:10.1002/9780470685983.

30 S. Bandyopadhyay and M. Bhattacharyya, “A Biologically Inspired Measure for Coexpression Analysis,” IEEE/ACM Trans. Computational Biology and Bioinformatics, vol. 8, no. 4, pp. 929-942, July/Aug. 2011.